

Technical Datasheet — VBF-T1R1FLASH-DSH-01

Next-generation BEV sedan cell · NMC811 / graphite+SiOx · Generation date: 2026-08-25

Field	Value	Source
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Rated capacity (Ah) 5.0 nominal (parameter set) / 4.936 verified (1C DFN) OKane2022 params;		
cell/final_1c_dfn.json:capacity_ah		
Nominal voltage / window (V) 3.56 midpoint / 2.5-4.2 cell/final_energy.json:midpoint_voltage_v; parameter set		
Rated energy (Wh) 17.51 (1C discharge integration) cell/final_energy.json:energy_wh		
Energy density (Wh/kg) 459.4 (contract caliber, electrolyte excluded) / 893.5 Wh/L cell/final_energy.json		
Maximum continuous discharge rate 1C verified (4.936 Ah delivered at 1C) cell/final_1c_dfn.json		
Fast-charge capability 4C @45 °C ambient: T_max 50.2 °C (≤60 °C), no plating (anode min +0.0114 V)		
cell/archA_h150_f4b_4c.json		
Required cooling liquid cooling $h \geq 120 \text{ W/m}^2 \cdot \text{K}$ (design point 150); without it T_max 94 °C at h=10 baseline_4c vs final_4c		
Operating temperature range Simulation conditions only: 25 °C (1C), 45 °C (4C); wider range not simulated –		
Cycle life Not simulated (aging protocol not in task scope); no fabricated value –		
Safety determination Overcharge to 4.7 V: thermal runaway NOT triggered (ODE verdict false); plating: none at 4C		
cell/final_tr.json:triggered; final_4c		
Dimensions & mass 1580 × 65 mm strip, stack 0.1908 mm; 38.11 g (electrolyte-excluded caliber) parameter set;		
cell/final_energy.json		
DC resistance 20.3 mΩ (10% discharge DCR) cell/final_energy.json:dcr_ohm		
Power density 5314 W/kg ($V_{OC}^2/4R \div \text{mass}$) cell/final_energy.json:power_density_w_kg		